APPROXIMATION FOR HENSELIAN PAIRS

THE STACKS PROJECT AUTHORS

Editorial note. The source compresses the Popescu application (obtaining b_i inside B with f_j(b) = 0) into one sentence; the colimit extraction of the solution system is carried as an explicit core obligation via the when-colimit framing of theorem-popescu.

Allowed standard prerequisites (editorial).

- Popescu's theorem. theorem-popescu (carded L26819). Factors the solution system through a smooth A-algebra inside the I-adic completion.
- Henselization of a G-ring pair is a G-ring. More on Algebra, lemma henselization-pair-G-ring (cross-chapter, named). Gives the reduction (3) implies (2).
- G-ring maps regularly to the completion. More on Algebra, lemma map-G-ring-to-completion-regular (cross-chapter, named). Gives the reduction (2) implies (1).
- Lifting a section along a smooth map. More on Algebra, lemma lift-section-smooth-morphism (cross-chapter, named). Lifts the map B -> A/I^N through an etale A -> A' with A/I^N isomorphic to A'/I^NA'.
- Retraction of etale maps over a henselian pair. More on Algebra, lemma characterize-henselian-pair (cross-chapter, named). Retracts A' back to A, bringing the solution into A.

Source reference. Original reference 09XI: focused-seeds/geometry/S9-smo-henselian-pair/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A02: focused-seeds/geometry/S9-smo-henselian-pair/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AH3: focused-seeds/geometry/S9-smo-henselian-pair/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07M7: focused-seeds/geometry/S9-smo-henselian-pair/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AH2: focused-seeds/geometry/S9-smo-henselian-pair/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 07GC: sources/stacks-smoothing/pinned-original-source.tex, source lines 3330–3330. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let (A, I) be a henselian pair with A Noetherian. Let $A^\wedge$ be the I-adic completion of A. Assume at least one of the following conditions holds*

(1) *$A \to A^\wedge$ is a regular ring map,*
(2) *A is a Noetherian G-ring, or*
(3) *(A, I) is the henselization (More on Algebra, Lemma 0A02) of a pair (B, J) where B is a Noetherian G-ring.*

Given $f_1, \ldots, f_m \in A[x_1, \ldots, x_n]$ and $\hat{a}_1, \ldots, \hat{a}_n \in A^\wedge$ such that $f_j(\hat{a}_1, \ldots, \hat{a}_n) = 0$ for $j = 1, \ldots, m$, for every $N \geq 1$ there exist $a_1, \ldots, a_n \in A$ such that $\hat{a}_i - a_i \in I^N$ and such that $f_j(a_1, \ldots, a_n) = 0$ for $j = 1, \ldots, m$.

Proof. By More on Algebra, Lemma 0AH3 we see that (3) implies (2). By More on Algebra, Lemma 0AH2 we see that (2) implies (1). Thus it suffices to prove the lemma in case $A \to A^\wedge$ is a regular ring map.

Let $\hat{a}_1, \ldots, \hat{a}_n$ be as in the statement of the lemma. By Theorem 07GC we can find a factorization $A \to B \to A^\wedge$ with $A \to P$ smooth and $b_1, \ldots, b_n \in B$ with $f_j(b_1, \ldots, b_n) = 0$ in B. Denote $\sigma : B \to A^\wedge \to A/I^N$ the composition. By More on Algebra, Lemma 07M7 we can find an étale ring map $A \to A'$ which induces an isomorphism $A/I^N \to A'/I^N A'$ and an A-algebra map $\tilde{\sigma} : B \to A'$ lifting σ. Since (A, I) is henselian, there is an A-algebra map $\chi : A' \to A$, see More on Algebra, Lemma 09XI. Then setting $a_i = \chi(\tilde{\sigma}(b_i))$ gives a solution. □